Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
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Project Title:	Mathematical Model for Scheduled Light Exposure: Circadian/Performance Countermeasure		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
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Human Research Program Risks:	None		
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Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kronauer, Richard (Harvard Medical School)		
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	The Specific Aims of our original proposal were: Specific Aim 1: To further develop and refine our 'Dynamic Stimulus Processing' Light Model so that it can accurately predict the phase and amplitude of the human circadian system under any lighting conditions, especially those that occur in space. This will be done using data from four completed studies of the effects on the human circadian system of: i) three-cycles of brief bright light pulses; ii) three-cycles of extended bright light pulses; iii) three-cycles of extended low- and moderate-intensity light pulses; and iv) single- and double-cycles of amplitude-suppressing critically-timed extended bright light pulses. Specific Aim 2: To validate the Light Model refined above in Specific Aim 1 using data from four other completed studies of the effects on the human circadian system of: v) single-cycle patterns of brief bright light pulses; vi)		

Task Description:	single-cycle extended bright light pulses; vii) single-cycle extended light pulses across a wide range of intensities; and viii) sleep-wake/light-dark schedules with a wide range of periods (11-h, 20-h, 23.5-h, 24-h, 24.6-h, 28-h, 42.85-h), different light intensities during wake (1, 8, or 15 lux), and with or without a single exposure to an extended bright light stimulus (in the 11-h condition only). Specific Aim 3: To incorporate the Light Model refined and validated above in Specific Aims 1 and 2 into our mathematical Neurobehavioral Performance Model, which will then be validated against experimental performance data collected under the wide variety of lighting conditions encompassed in the eight studies described above in Specific Aims 1 and 2. Specific Aim 4: To develop a user-friendly Circadian Performance Simulation Software (CPSS) package that can be used to specify appropriate light schedules as a countermeasure to the poor performance and sleep quality associated with circadian misalignment in space. The key findings of the project are: 1) We refined and validated our current circadian pacemaker model to predict the effects of different stimuli on entrainment and phase resetting of pacemaker at low light levels 2) We characterized the amplitude recovery dynamics of the endogenous pacemaker as well as identified different excitation regions of the pacemaker 3) Our user-friendly Circadian Performance Simulation Software (CPSS) package was tested with actual mission schedule to evaluation our predictions of performance and alertness during pre and post launch conditions. 4) A preliminary algorithm for schedule assessment and countermeasure design was developed based on the analysis of different schedules using CPSS The impacts of these findings are: 1) The impact of finding 1 above suggests that the circadian pacemaker can be entrained using different stimuli and the current mathematical model can be used for simulating the effect of light on pacemaker dynamics during long duration space missions 2) The impact of finding 2 above suggests that the rate of recovery from amplitude reduction is slower and therefore returns to equilibrium conditions make take longer. This has implications for the rate of recovery after disruption of the pacemaker (e.g. by different shift schedules or lighting conditions). Amplitude reduction of the pacemaker may affect the influence of circadian pacemaker on sleep consolidation or neurobehavioral performance during the daytime. 3) The impact of finding 3 is that there is now available a software package to evaluate our predictions of the performance and alertness of crewmembers during long-duration space missions. 4) The impact of finding 4 is that the algorithm can be incorporated into our software package to assess the schedules of long duration space missions in order to design appropriate countermeasures to improve the performance and alertness of crewmembers on these missions. This is the final progress report. There are no plans on this project for the coming year except continued publication and presentation of findings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research focuses on the further development of mathematical models and software that aid in schedule design to improve performance (and thereby public safety) for people who work at night, on rotating schedules or on non-24-hour schedules (pilots, train and truck drivers, shift workers, health care workers, etc.). This research also aids in the specification of lighting requirements aboard spacecraft and in other work conditions to insure proper entrainment and circadian phase of these workers, even if they work at night or on rotating schedules.
Task Progress:	1. Specific aims 1-4 are complete. 2. For Specific aim 4, our user friendly software – Circadian Performance Simulation Software (CPSS) - was further tested with actual mission schedules to evaluation our predictions of performance and alertness during pre and post launch conditions. 3. We added a new component to our current version of the model to predict the response of the core pacemaker at low light levels due to nonphotic stimulus. 4. We also characterized the amplitude recovery dynamics of the circadian pacemaker.
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